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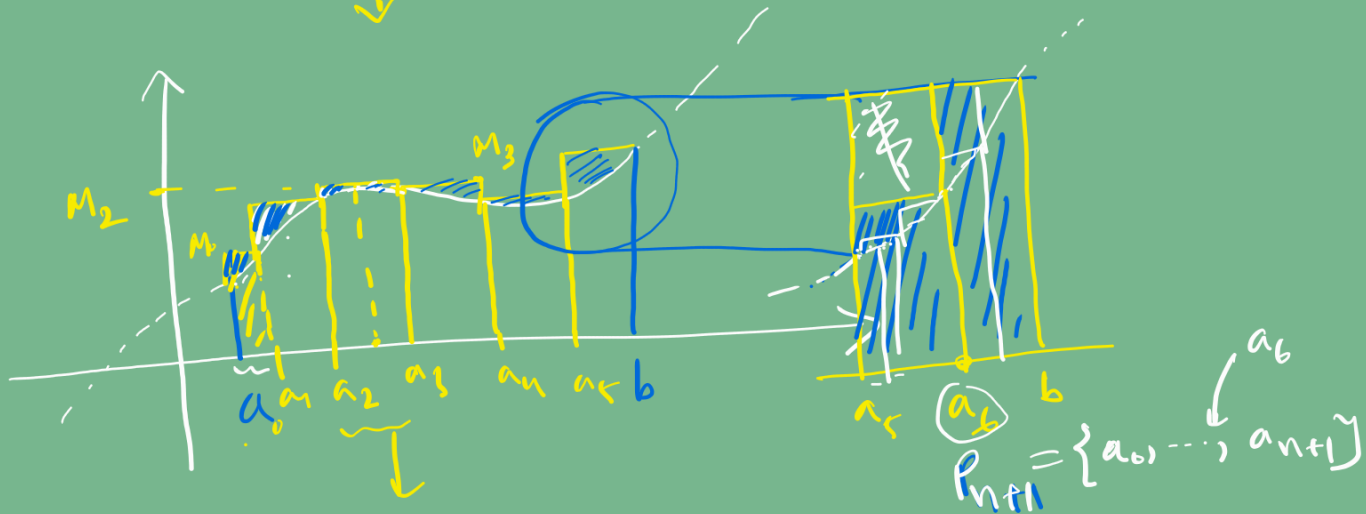
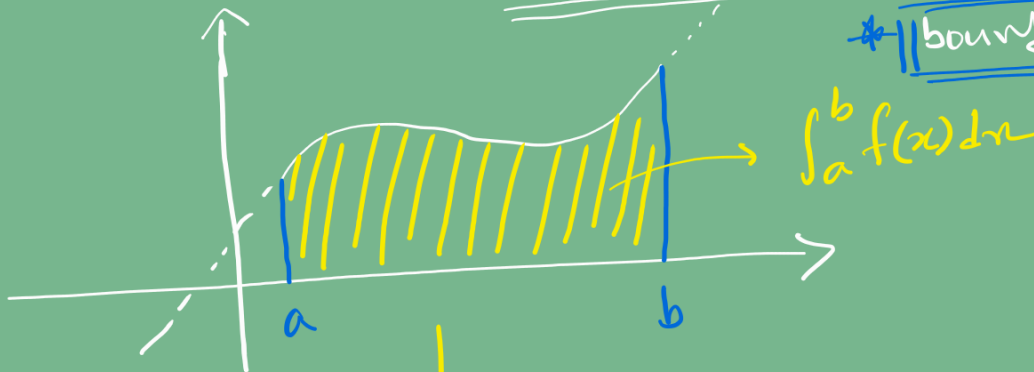
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• Necessary condⁿ: $f: [a, b] \rightarrow \mathbb{R}$

* bounded



$$M_0(a_1 - a_0) + M_1(a_2 - a_1) + M_2(a_3 - a_2) + \dots$$

$$= \sum_{i=0}^{n-1} M_i(a_{i+1} - a_i) = U(f; P_n), P_n = \{a_0, a_1, \dots, a_n\}$$

$$\|P_n\| = \max_{0 \leq i \leq n-1} \{ (a_{i+1} - a_i) \}$$

$$\lim_{n \rightarrow \infty} \|P_n\| = 0 ; \quad U(P_n, f) > U(P_{n+1}, f) > \dots$$

$$L(P_n, f) < L(P_{n+1}, f) < \dots$$

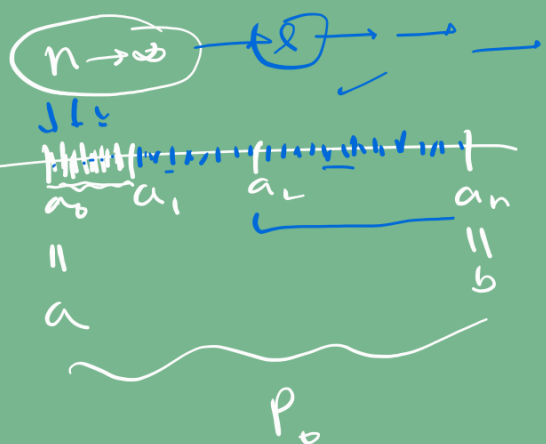
$$\underbrace{U(P_0, f) > U(P_1, f) > \dots}_{n \rightarrow \infty} \quad \underbrace{L(P_0, f) < L(P_1, f) < \dots}_{n \rightarrow \infty}$$

$\int_a^b f(x) dx$
 $= \lim_{n \rightarrow \infty} U(P_n, f)$
 $= \lim_{n \rightarrow \infty} L(P_n, f)$

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Note: $\lim_{n \rightarrow \infty} U(P_n, f)$ & $\lim_{n \rightarrow \infty} L(P_n, f)$ always exist,
 if f is bdd. [provided $\lim_{n \rightarrow \infty} \|P_n\| = 0$]

$$\lim_{n \rightarrow \infty} \|P_n\| = 0$$



$$\{U(P_n, f)\}_{n \in \mathbb{N}}$$

$\downarrow$
 $U(P_n, f)$

Monotone and bdd

Monotonically decreasing bounds below

$$\underbrace{U(p_n, f)}_{\text{decreasing}} > \underbrace{L(p_n, f)}_{\text{increasing}} > \underbrace{L(p_0, f)}_{\text{increasing}}$$



$$p_0 = \{ \dots \}$$

$$\sum \textcircled{0} < \infty$$

$$L(p_n, f) \uparrow$$

$$L(p_n, f) < U(p_0, f) < \infty$$



$$\lim_{\|p_n\| \rightarrow 0} U(p_n, f) = \int_a^b f(x) dx = \inf U(p_n, f)$$

$$\lim_{\|p_n\| \rightarrow 0} L(p_n, f) = \int_a^b f(x) dx = \sup L(p_n, f)$$

$$\|p_n\| = \textcircled{0} \quad \forall n \in \mathbb{N}$$

$$n \rightarrow \infty$$



$$\textcircled{5} \quad \text{If } U(p_n, f) \rightarrow \inf_{n \in \mathbb{N}} U(p_n, f) = \int_a^b f(x) dx,$$

$\|P_n\| \rightarrow 0$ and converse is also true.

$$\underbrace{U(P_n, f) - L(P_n, f)} < \varepsilon$$

Ex: A bounded function but not Riemann Integral

$$f: [0,1] \rightarrow \mathbb{R}$$

(*) $f(x) = \begin{cases} 1, & x \in \mathbb{Q} \cap [0,1] \\ 0, & x \in \mathbb{Q}^c \cap [0,1] \end{cases}$

$$\lim_{\|P_n\| \rightarrow 0} U(P_n, f) = 1 = \int_0^1 f$$

$$\lim_{\|P_n\| \rightarrow 0}$$

$$\lim_{\|P_n\| \rightarrow 0} L(P_n, f) = 0 = \int_0^1 f$$

$$\lim_{\|P_n\| \rightarrow 0}$$

Lebesgue Integral